

Quick Check

Lesson 4: The Hidden Link: Discovering $\tan \theta = \sin \theta / \cos \theta$

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.4: Trigonometry 1 | Lesson 4

Learners divided $\sin \theta$ by $\cos \theta$ for $\theta = 20^\circ, 30^\circ, 40^\circ, 50^\circ, 60^\circ$ and 70° , and compared each answer with the $\tan \theta$ value in their table. Every row matched, apart from tiny rounding differences. What is the best conclusion from this pattern?

- A** $\sin \theta \div \cos \theta$ is never equal to $\tan \theta$, but the values are close
- B** $\sin \theta \div \cos \theta$ is sometimes equal to $\tan \theta$, depending on the angle
- C** $\sin \theta \div \cos \theta$ is equal to $\tan \theta$ for every acute angle tested, so it looks like an identity
- D** $\sin \theta \div \cos \theta$ is equal to $\cos \theta \div \sin \theta$ for these angles

In a right-angled triangle, $\sin \theta = O/H$ and $\cos \theta = A/H$, where O is the opposite side, A the adjacent side and H the hypotenuse. What is $(O/H) \div (A/H)$ after simplifying?

A O/A

B $(O \times A)/H$

C A/O

D O/H

A Kenya Power engineer's calculator has a broken tan button. Her notebook says $\sin 50^\circ = 0.766$ and $\cos 50^\circ = 0.643$. Using $\tan \theta = \sin \theta \div \cos \theta$, what is $\tan 50^\circ$ to 3 decimal places?

A 0.839

B 1.191

C 1.409

D 0.492

The angle of elevation to the top of a mast is 50° , and the horizontal distance from the observer to the base is 40 m. Using $\tan 50^\circ = \sin 50^\circ \div \cos 50^\circ = 0.766 \div 0.643$ and height = $\tan \theta \times \text{horizontal distance}$, what is the height of the mast to 1 decimal place?

A 47.7 m

B 25.7 m

C 33.6 m

D 30.6 m

θ is an acute angle in a right-angled triangle and $\sin \theta = 3/5$. Without a calculator, what are $\cos \theta$ and $\tan \theta$?

A $\cos \theta = 4/5$ and $\tan \theta = 3/4$

B $\cos \theta = 4/5$ and $\tan \theta = 4/3$

C $\cos \theta = 2/5$ and $\tan \theta = 3/2$

D $\cos \theta = 3/5$ and $\tan \theta = 1$

For an angle of 35° , $\tan 35^\circ = 0.700$ and $\cos 35^\circ = 0.819$. Using $\tan \theta = \sin \theta \div \cos \theta$, what is $\sin 35^\circ$ to 3 decimal places?

A 1.519

B 0.855

C 0.573

D 1.170

A mast is 95.4 m tall, and an observer stands 40 m from its base. Using $\tan \theta = \text{height} \div \text{distance}$, what is $\tan \theta$ for the angle of elevation θ to the top?

A 2.385

B 135.4

C 55.4

D 0.419

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).